Mohamed Yousry

Embedded Software & Cybersecurity Engineer | Alexandria, Egypt | Open to Relocation
+201121670202 | Military Status: Completed
moh.yousry01@gmail.com linkedin.com/in/yousry01

Summary

Embedded Software & Cybersecurity Engineer with 4 years of experience developing safety- and security-critical embedded systems using C / Embedded C on NXP Power Architecture and ARM Cortex-M4 platforms. Hands-on embedded cybersecurity experience in HSM-backed secure boot, hardware root of trust, firmware authentication and integrity, secure firmware update, secure key storage, key management, AES-128, AES-CMAC, SHA-256, ECC P-256, ECDH, ECDSA, TARA, cybersecurity requirements, security architecture, replay protection, security verification and traceability aligned with IEC 62304 Class C and IEC 81001-5-1:2022.

Work Experience

Job Title: Senior Embedded Software Engineer

Company: Ezz Steel - Ezzmedical
Location: Giza, Egypt
Dates: Oct 2022 - Present
Medical device projects: ICU ventilator; blower-based ventilator; CPAP machine; syringe pump
Key Responsibilities and Achievements:
Embedded Cybersecurity Engineering

- Participate in medical device cybersecurity lifecycle activities including Item Definition, asset identification, Threat Analysis and Risk Assessment (TARA), security objectives and cybersecurity requirements.
- Define embedded security architecture and product-security mitigations for safety-critical device functions, including embedded security threats, attack vectors and mitigation controls.
- Implement cryptographic protections using AES-128 encryption, AES-CMAC message authentication, SHA-256 hashing, ECC P-256, ECDH and ECDSA where required by the security design.
- Design secure communication behavior with authentication, integrity checks and replay-attack mitigation.
- Support firmware authenticity and integrity verification across secure boot, secure firmware update and application authentication flows.
- Integrate HSM-backed cryptographic services, secure key storage, key management and key provisioning scripts for hardware-backed embedded security.
- Perform security verification and maintain requirements-to-mitigation traceability across cybersecurity requirements, controls and test evidence.

Hardware Security, Secure Boot & Firmware Update

- Design hardware root of trust on NXP Power Architecture by integrating HSM-backed cryptographic services and SHE secure boot behavior.
- Implement firmware authenticity and integrity verification using AES-CMAC application authentication metadata in secure boot and secure firmware-update images.
- Implement secure key storage, key-management scripts and key-provisioning flows following the SHE key storage protocol.
- Design bootloader/application memory partitioning and secure update sequence, including OTA firmware update and application authentication before activation.
- Configure JTAG password protection, secure debug access and flashing controls to reduce unauthorized debug and firmware access.

Embedded Firmware Development & Device Integration

- Develop real-time, safety-critical firmware for Class C medical devices from software and low-level requirements.
- Build device-level firmware features that connect MCAL/HAL drivers, middleware and application logic for ventilation and infusion devices.
- Implement C / Embedded C firmware on industrial microcontrollers including NXP Power Architecture and ARM Cortex-M4-based platforms.
- Integrate NXP SDK drivers including ADC, GPIO, C55Flash, Security library, CAN, PWM, I2C and UART.

Safety, Architecture & Process Engineering

- Develop software following SDLC and compliance expectations including IEC 62304 Class C, IEC 81001-5-1:2022, ISO 13485, ISO 14971, MISRA-C and FDA guidance; analyze safety and cybersecurity risks and translate hazards, threats and mitigations into embedded requirements.
- Design layered firmware architecture across MCAL, HAL, Middleware and Application Layer with state machines, device drivers and fail-safe communication behavior.

Verification, Testing & Traceability

- Perform unit and integration testing using VectorCAST, functional testing, SIL/HIL verification, security verification, traceability and test reporting.
- Create and maintain SOPs and project documentation for configuration management, Secure SDLC, firmware release and change control.
- Contribute to software design documentation, test artifacts, release documentation, design reviews and audits.

Real-Time Control & Algorithm Optimization

- Design and tune control algorithms (PID, PI, ADRC) for pressure and flow control, reducing controller rise time from 1000 ms to 80 ms.

Improve sensing accuracy via calibration techniques, digital filtering and validation methods.

- Implement real-time signal processing, sensor interfacing, diagnostics and configuration handling between controller, sensor and actuator modules.
- Implement RTOS applications using tasks, queues, semaphores, mutexes, timers and interrupt synchronization.

Earlier Technical Experience

Founder / Technical Lead - CompuTech; Partner & Head of Maintenance - GEEK | Alexandria, Egypt | 2017-2022

- Led laptop/computer maintenance operations across both businesses, covering hardware diagnostics, repair workflow, quality checks, sourcing, customer support and issue resolution.

Technical Skills

- Embedded Cybersecurity: Secure Boot, HSM, Hardware Root of Trust, SHE, Secure Firmware Update, Firmware Authentication & Integrity, Secure Key Storage, Key Management, Key Provisioning, JTAG/Secure Debug Protection, TARA, Replay Protection, Security Mitigations, Security Verification.
- Cryptography: AES-128, AES-CMAC, SHA-256, ECC P-256, ECDH, ECDSA, Cryptographic RNG.
- Embedded Software: C, Embedded C, Bare-Metal, RTOS, MCAL, HAL, Middleware, Device Drivers, Bootloaders, Startup Code, Flash/Memory Management, NXP Power Architecture, ARM Cortex-M4.
- Communication Protocols: CAN, SPI, I2C, UART.
- Safety & Cybersecurity: IEC 62304 Class C, IEC 81001-5-1:2022, ISO 13485, ISO 14971, MISRA-C, FDA Guidance, Secure SDLC, Cybersecurity Risk Management.
- Testing & Verification: VectorCAST, CANoe, Unit Testing, Integration Testing, Functional Testing, SIL/HIL, Security Verification, Requirements Traceability.
- Control Systems: PID, PI, ADRC, Real-Time Signal Processing, Sensor Calibration, Digital Filtering.

Tools and IDEs

- IDEs & Toolchains: S32 Design Studio, NXP SDK, STM32Cube, Eclipse, Atmel Studio, startup, flash and MCU configuration, pre-build/post-build scripts.
- Debugging & Flashing Tools: P&E Micro debuggers, JTAG password protection, secure/debug access and flash programming.
- Version Control: Git/GitHub, configuration management and change control.
- Documentation & Modeling: Enterprise Architect, Draw.io, Microsoft Visio (architecture, software design, safety, cybersecurity requirements and test artifacts).

Education

- Bachelor's Degree in Mechanical Engineering - Alexandria University, Jul 2017; NTI DEY Diploma in Embedded Systems - Apr 2021

Certifications & Courses

- CompTIA Security+ - In progress; IMT Diploma - Embedded software ARM, AVR, RTOS, Automotive Communication; ISO/IEC 17025:2017 Training - NIS; Sprints ISTQB Master Class - Foundation Level, Agile

Languages

- English - Professional Proficiency; Arabic - Native